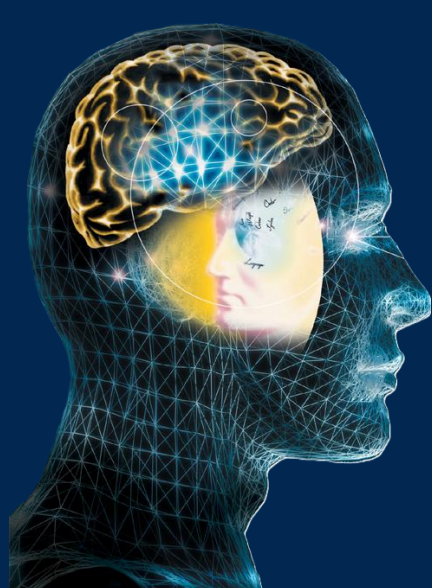




The development of navigation and attentional abilities shapes how children explore naturalistic environments

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THE QUESTION

Children are often called “*natural explorers*,” but most evidence comes from small, simple lab settings. How do children explore **large, naturalistic environments** when they must **navigate** and **manage attention** at the same time?

APPROACH

- 42 children (7–14 yrs) and 42 adults (18–23 yrs)
- Free exploration of a **4,216 m² immersive VR furniture store** (23 rooms, 355 objects, 10 surprising multisensory events)
- Eye, head, and body tracked throughout two 4-min exploration blocks
- Followed by a 10-trial visual search task.

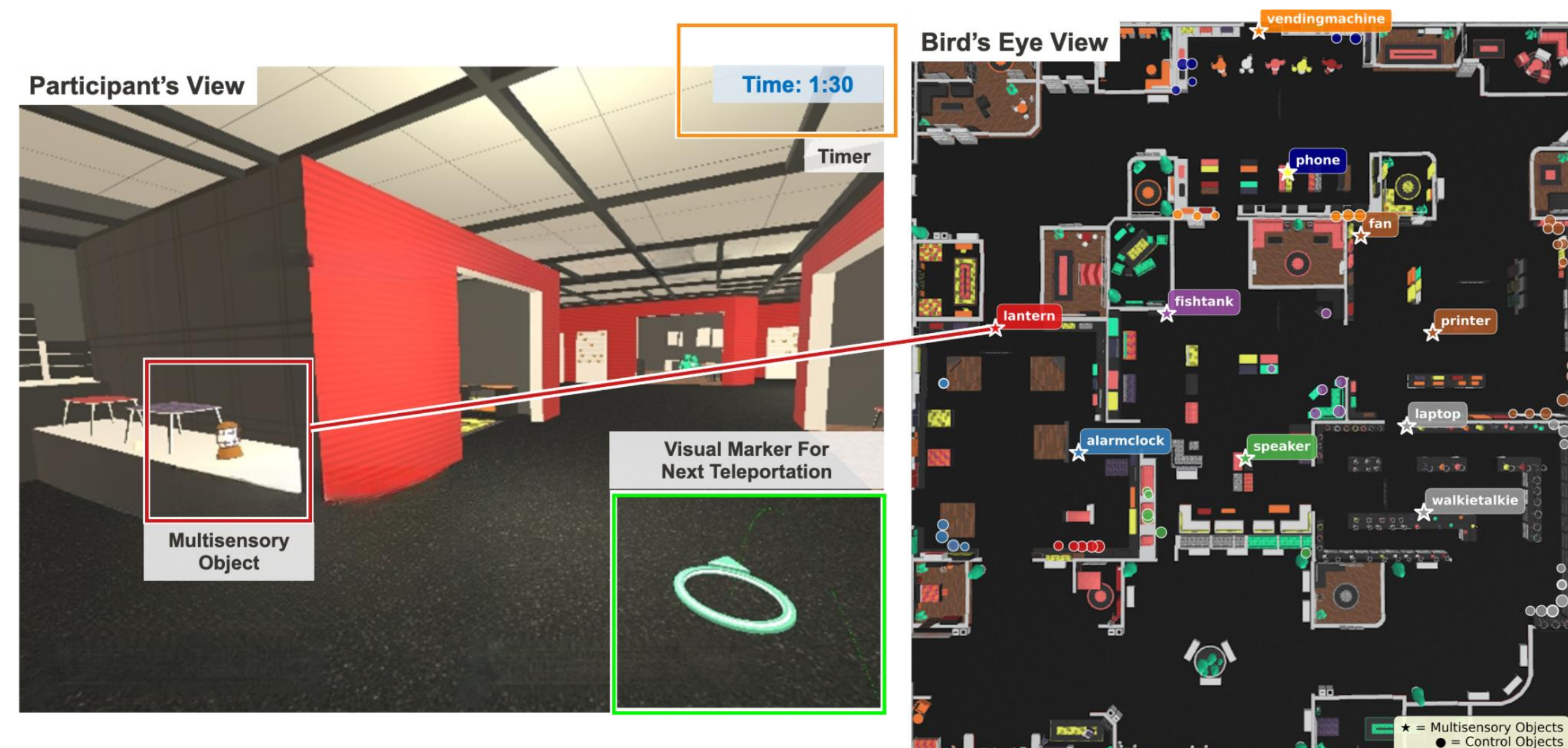


Fig. 1. First-person view (left) and bird's-eye map (right) of the VR store. Stars mark the 10 multisensory objects.

TWO BEHAVIORAL SIGNATURES

We isolated two behaviors that index **navigation** and **attention** during free exploration:

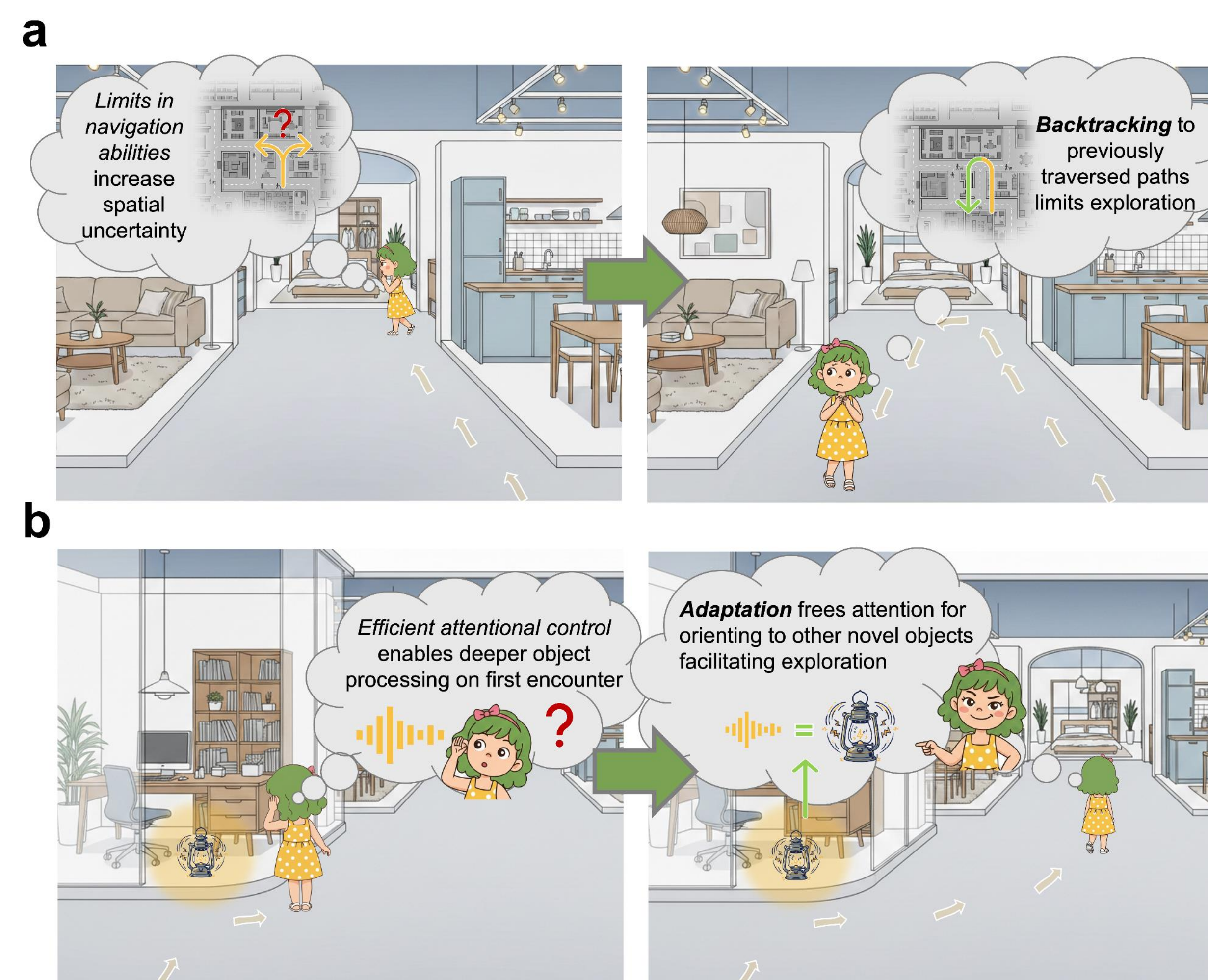


Fig. 2. (a) **Backtracking duration** — time spent retracing previous paths when uncertain about location. (b) **Adaptation to multisensory events** — how quickly stop durations decrease across repeated encounters with the same salient object.

KEY FINDINGS

1 Global Exploration

Children covered **less space** and looked at **fewer unique objects** than adults — even though they walked the **same total distance**.

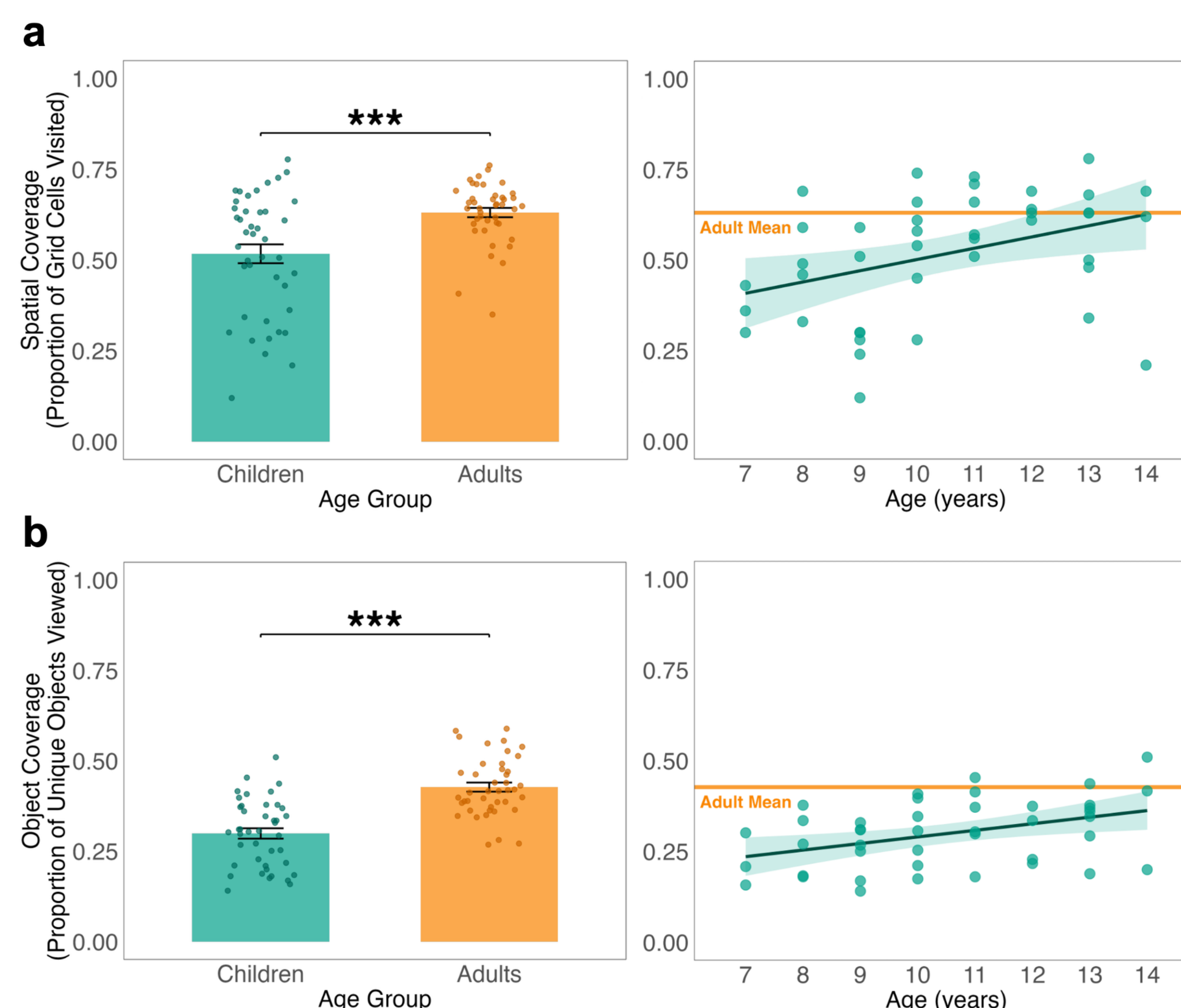


Fig. 3. Spatial and object coverage by group (left) and across age (right). *** $p < .001$.

3 Children adapt slowly

Adults stop **longer first**, then **adapt fast**. Children keep stopping at every encounter — **limiting what they sample**.

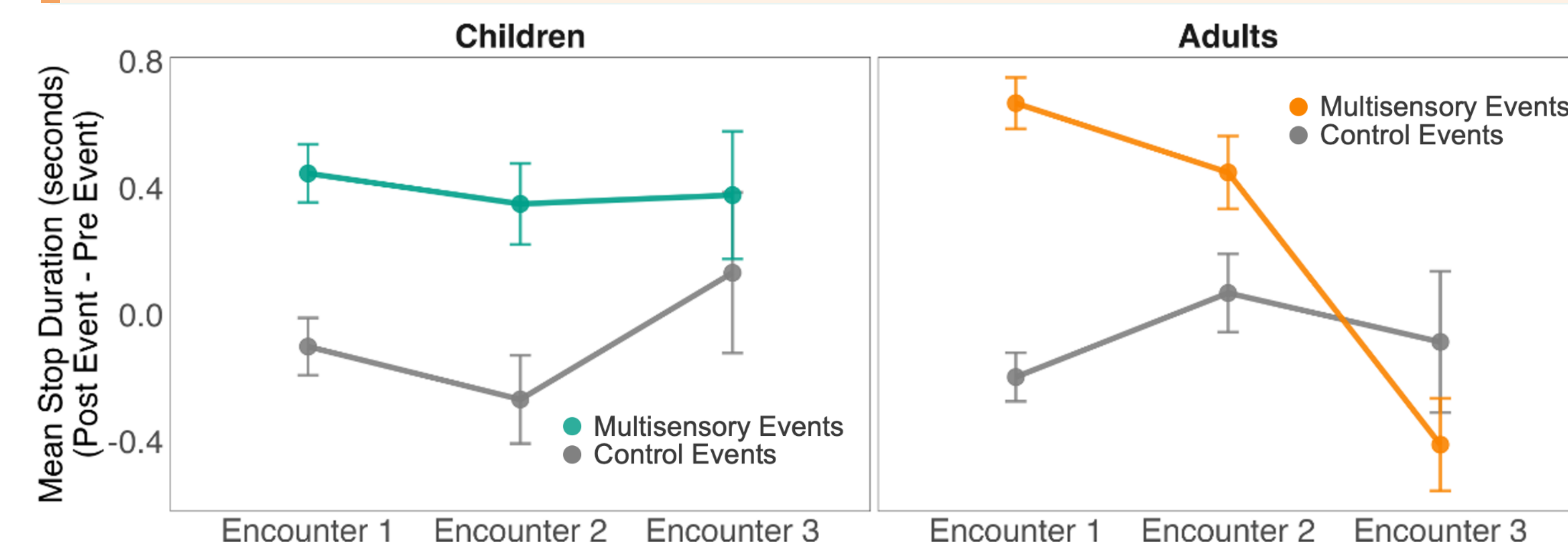


Fig. 5. Stop duration after multisensory vs. control events across 3 encounters.

2 Children backtrack longer

Same backtrack **distance** as adults, but **much longer time** — a sign of **spatial confusion**, not strategy.

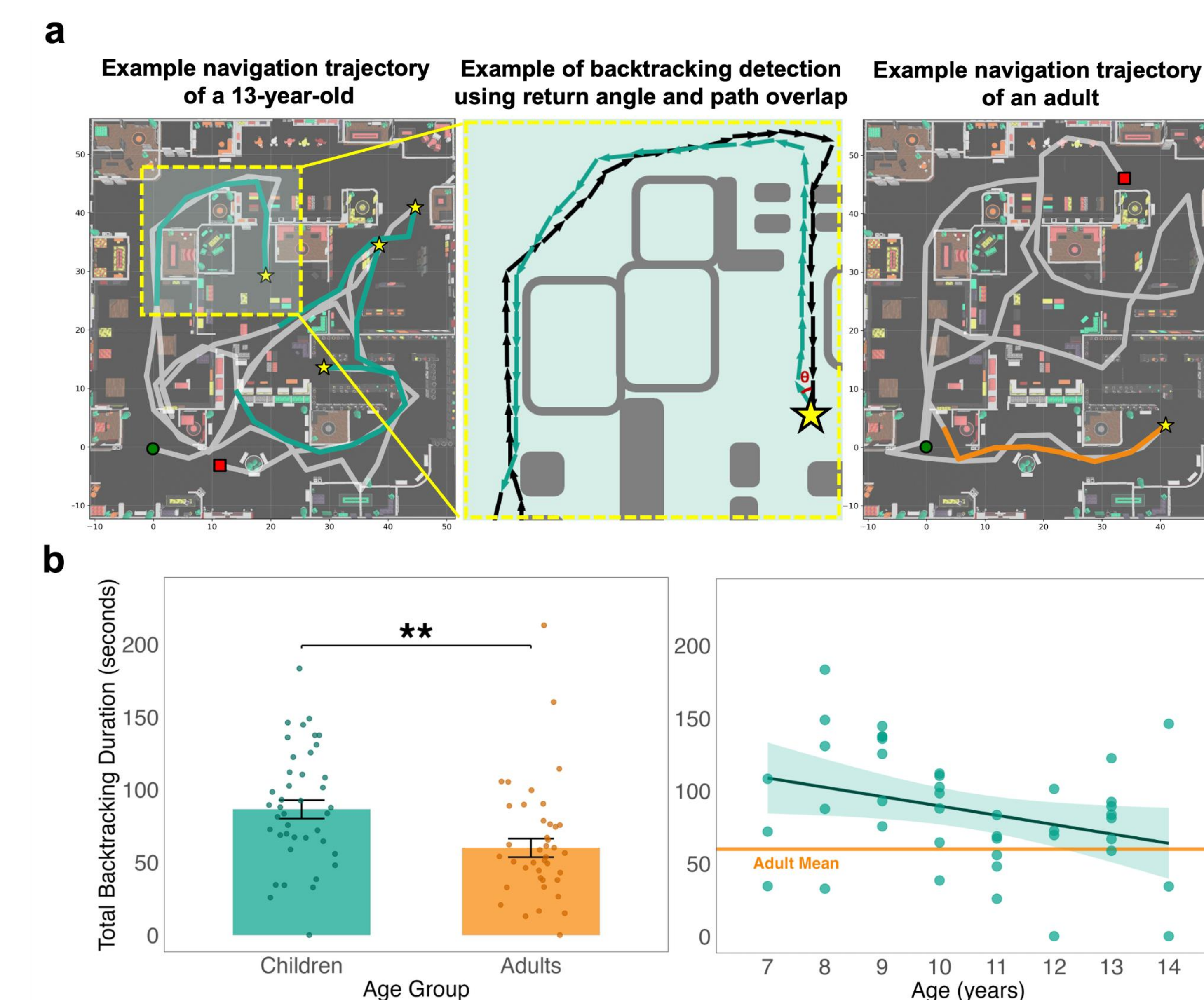


Fig. 4. (a) Trajectories + detection. (b) Backtracking duration by group/age. ** $p < .01$.

Backtracking longer -> poor spatial learning

After exploring, participants searched for **10 target objects**. Children who **backtracked more** during exploration found **fewer targets later** — linking moment-to-moment navigation to **lasting spatial memory**.

TAKEAWAY

- Children explore **less comprehensively** than adult, not because they **move less**, but because **developing navigation and attention systems** constrain how effectively they can do both at once.
- These constraints only emerge in **large, sensory-rich environments**, the kind that controlled lab tasks miss.
- How children explore depends on the **coordination** of developing cognitive systems, itself a developmental achievement.